Disturbance and resilience of Europe‘s forests

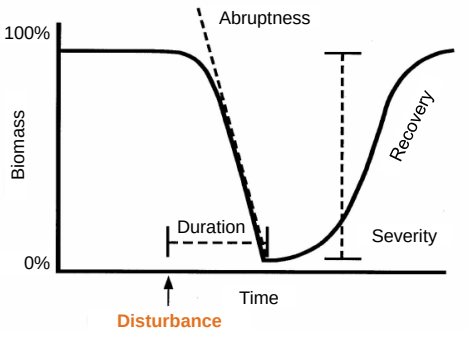
Rupert Seidl

Acknowledgements

A wonderful team
Inspiring (inter)national collaborators
Supportive funding institutions

Disturbance

Definition: discrete events in space and time that lead to the loss of live biomass and alter the resource availability in ecosystems and communities.



White und Jentsch (2001), Jentsch et al. (2019)

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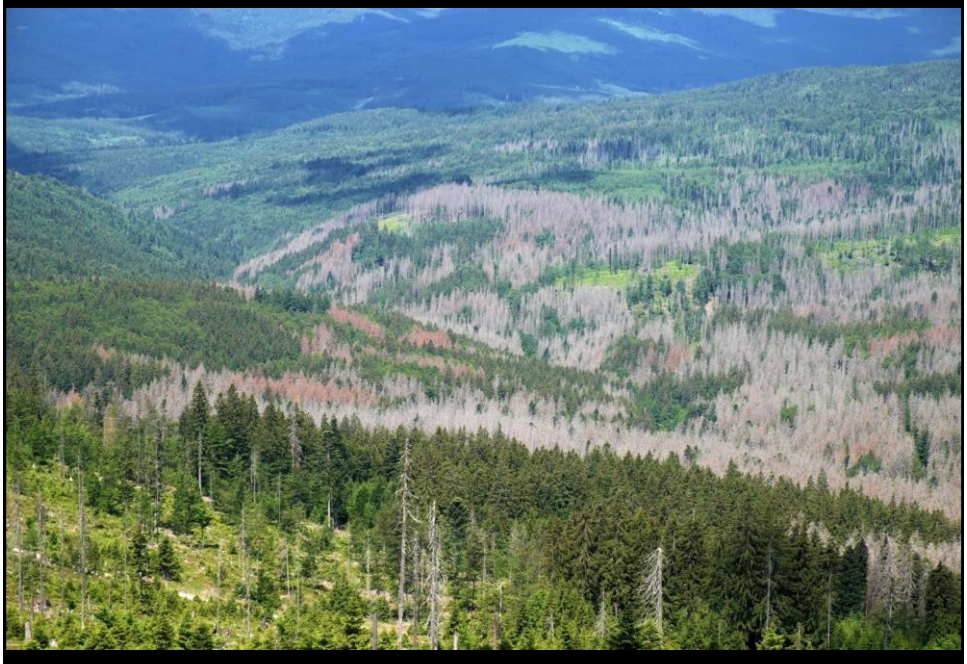
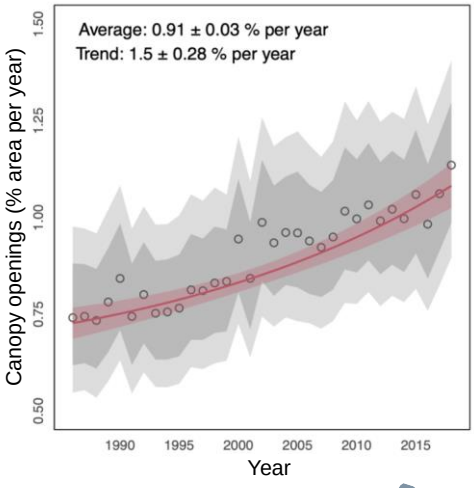


Image credit: Cameron Strandberg (CC)

Disturbances in Europe are increasing

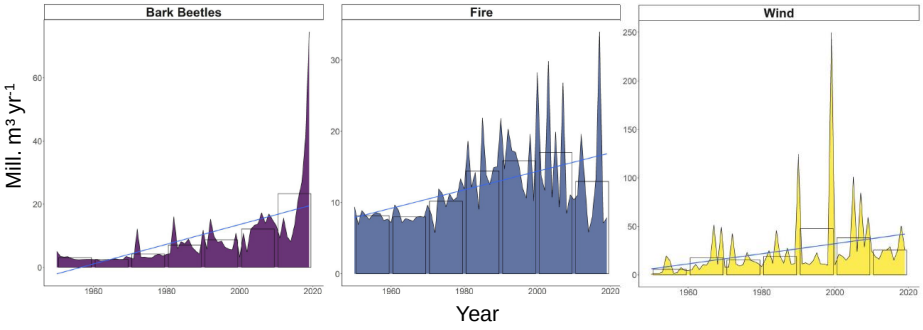
Canopy openings increase by 1.50% per year

In Central Europe, disturbance rates have more than doubled since the mid 1980s



Senf et al. (2018, Nature Comm.), Senf et al. (2021, One Earth)

Natural disturbances in Europe



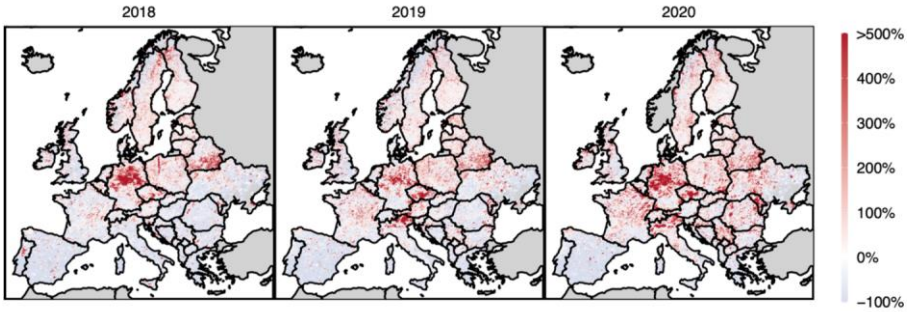
All natural reasons of tree mortality are increasing

Biggest increase for bark beetles, more than doubling in the past 20 years

Patacca et al. (2022, Glob. Change Biol.)

Excess canopy mortality 2018-2020

The 2018-2020 drought caused the biggest wave of tree mortality in Europe in at least 170 years



Excess canopy mortality relative to 1986-2015

Senf and Seidl (2021, Biogeosciences)

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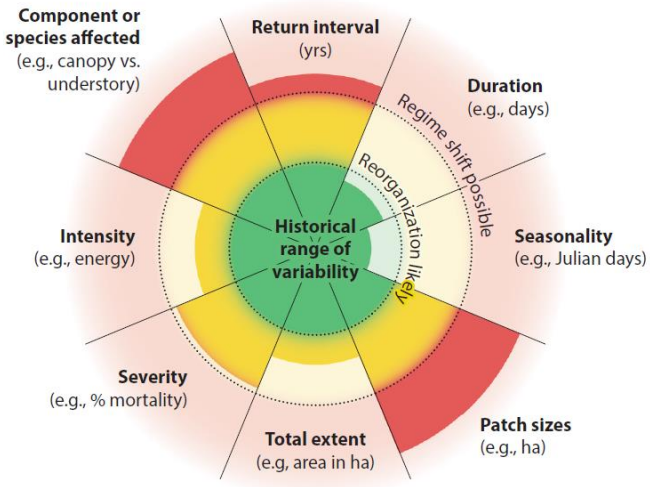
Emergence of novel disturbance regimes



Turner and Seidl (2023, AREES)

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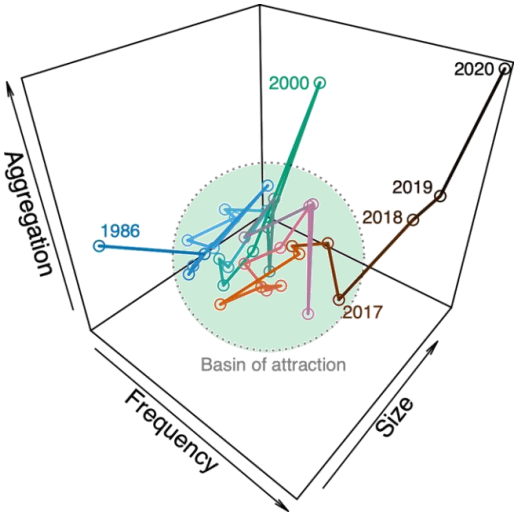
Emergence of novel disturbance regimes



The European forest disturbance regime of the years 2018-2020 has no analogy in the recent past

Senf and Seidl (2021, Biogeosciences)

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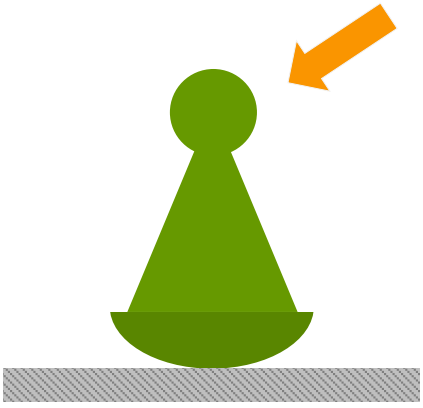
How resilient are Europe's forests

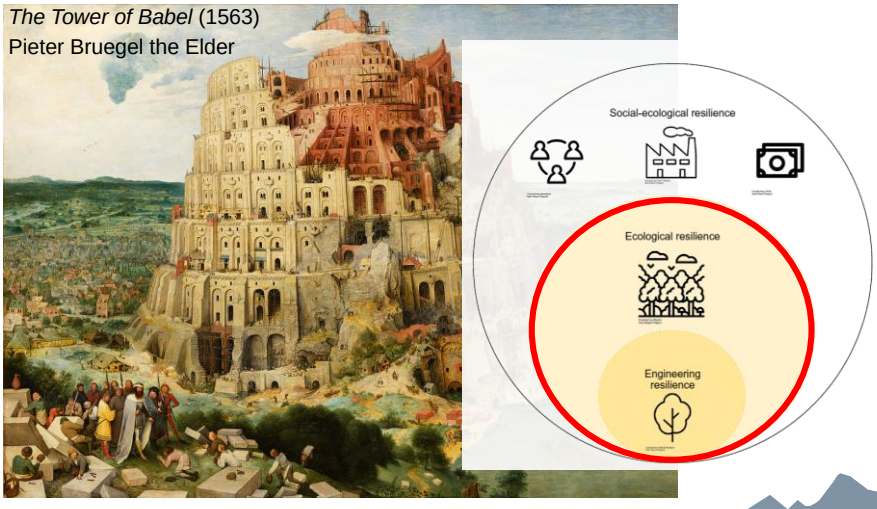
Resilienz

The ability of a system to return to its pre-disturbance state or a functionally equivalent state governed by the same drivers and feedbacks



Holling (1973, AREES), Nikinmaa et al. (2020, Curr. For. Rep.)

Resilience



Nikinmaa et al. (2020, Curr. For. Rep.)

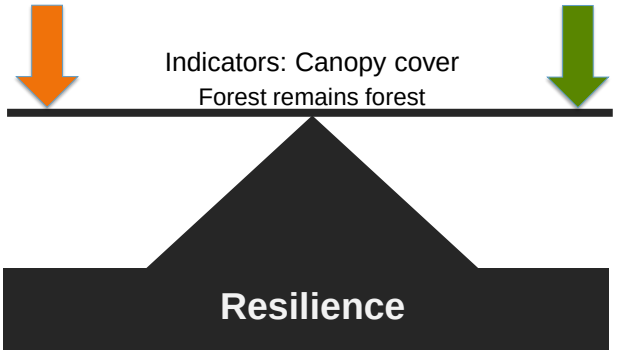
How resilient are Europe‘s forests

Time between two disturbances

Disturbance intervall

Time until recovery after disturbance

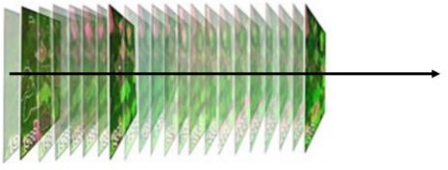
Recovery intervall

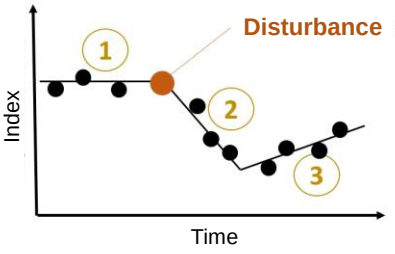


Senf and Seidl (2022, Glob. Ecol. Biogeogr.)

Analysis of Landsat satellite images

Annual composits of cloud-free images



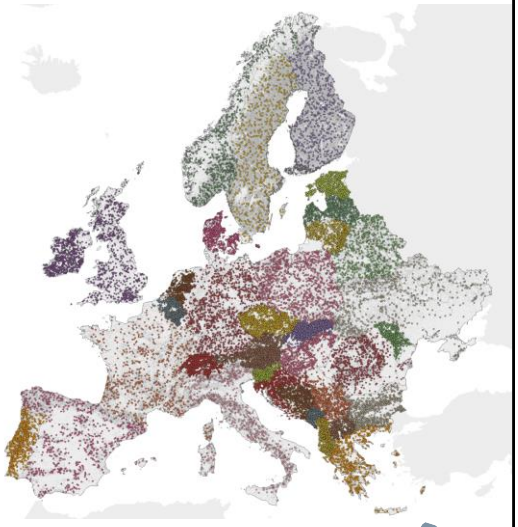


Cohen et al. (2010, Rem. Sens. Environ.), Senf and Seidl (2021)

Reference data

Manual interpretation of
~ 680,000 image chips for
~ 20,000 sites across Europe

Spatial & temporal coverage
1986-2020
210 Mill. ha of forest
35 countries

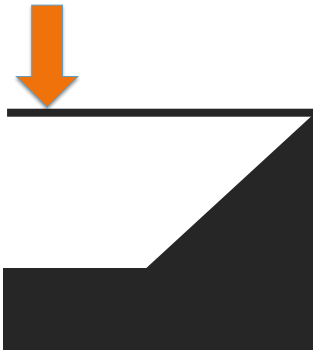


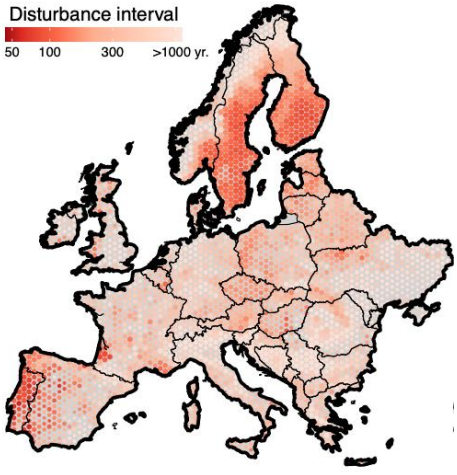
Senf and Seidl (2021, Nature Sust.)

Disturbance intervall

Time between two disturbances

Disturbance intervall





Senf and Seidl (2022, Glob. Ecol. Biogeogr.)

How resilient are Europe‘s forests

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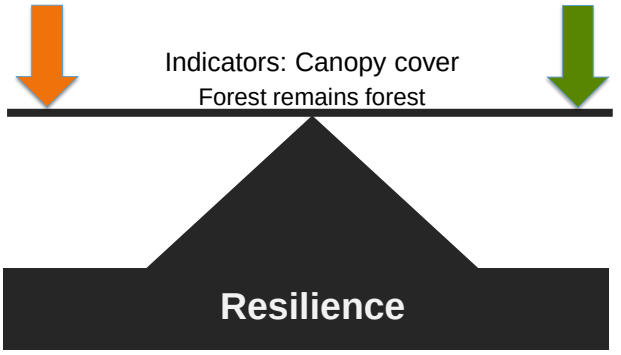
Time until recovery after disturbance

Recovery intervall



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Estimation of recovery

Estimate canopy cover

Based on reference data from 11,221 high resolution aerial images and a random forest model using spectral bands and indices as predictors

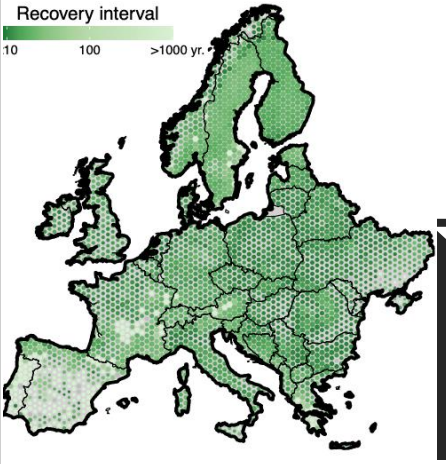
Derive recovery relative to pre-disturbance state

Defined as average canopy cover five years before a disturbance

Calculate recovery interval from space-for-time substitution

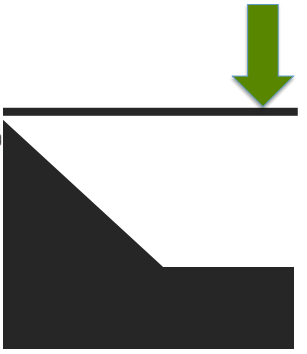
Logisitc model over all recovering disturbances in a cell

Recovery intervall



Time until recovery after disturbance

Recovery intervall



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How resilient are Europe‘s forests

Time between two disturbances

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Indicators: Canopy cover
Forest remains forest

Senf and Seidl (2022, Glob. Ecol. Biogeogr.)

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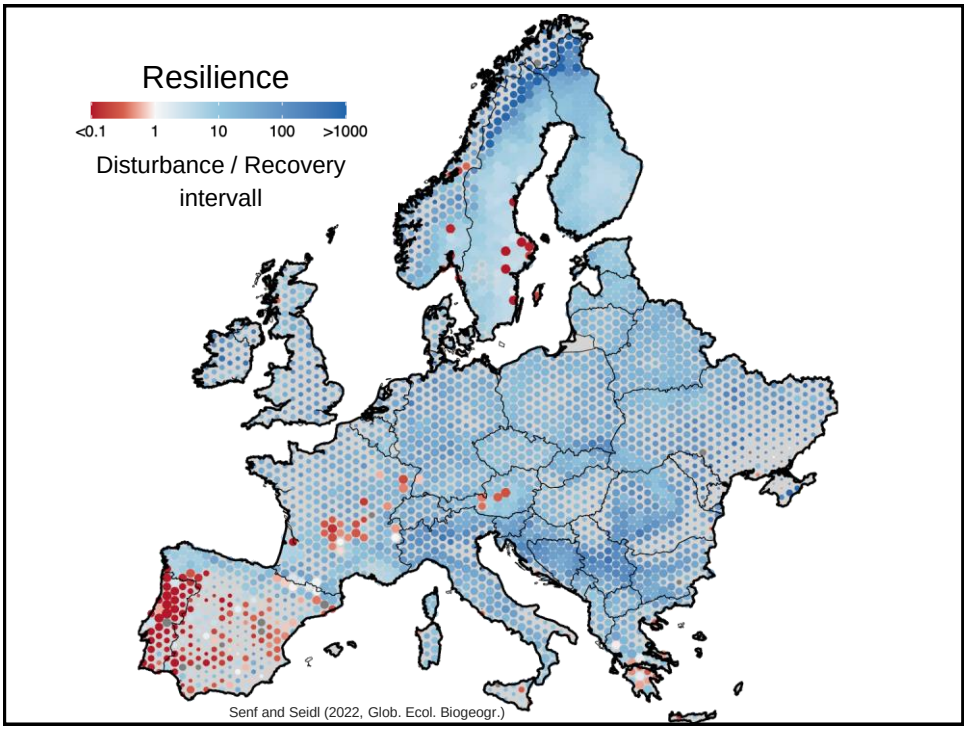


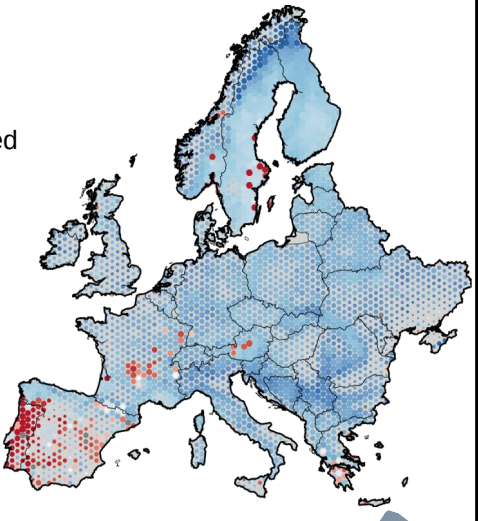
How resilient are Europe‘s forests

Resilience generally high

69% of Europe’s forest area recovers more than 10x faster than it is disturbed

But…

14% of Europe’s forests have low resilience, with disturbance intervals equal to or shorter than recovery intervals



Resilience

low high

<0.1 1 10 100 >1000

Disturbance interval / Recovery interval

Senf and Seidl (2022, Glob. Ecol. Biogeogr.)

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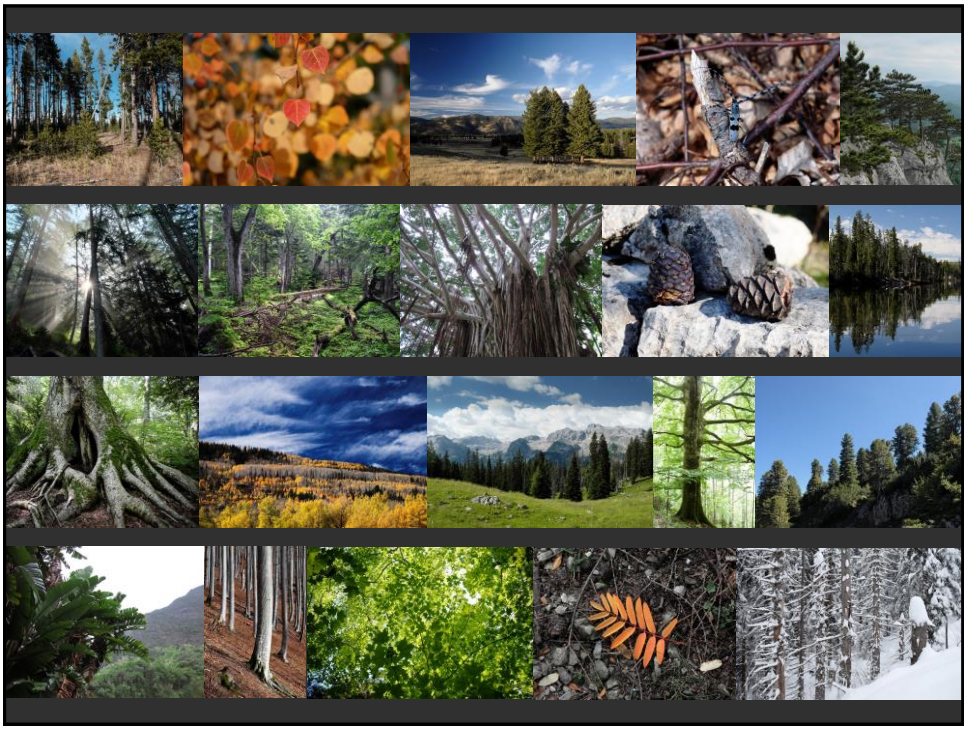
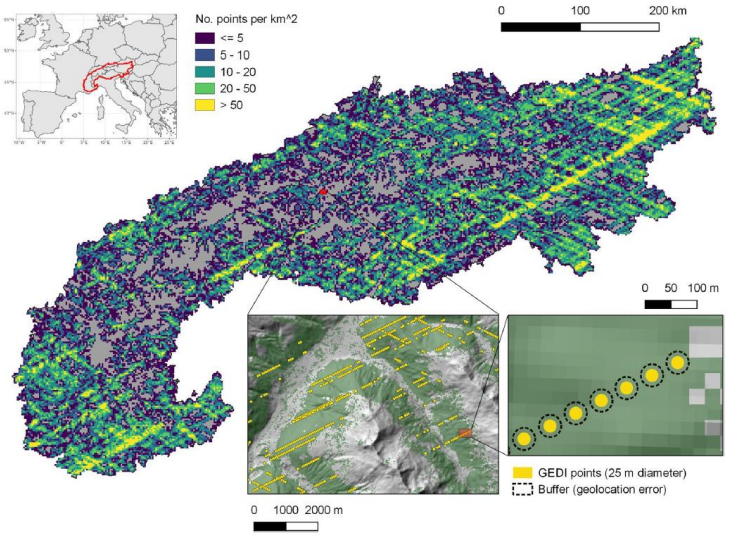
Recovery beyond forest cover…

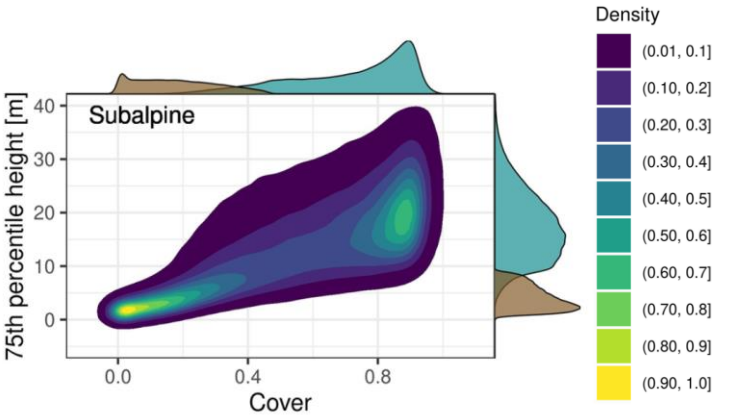


Recovery of forest structure



Stritih et al. (2023, Landscape Ecol.)

Structural recovery across the Alps



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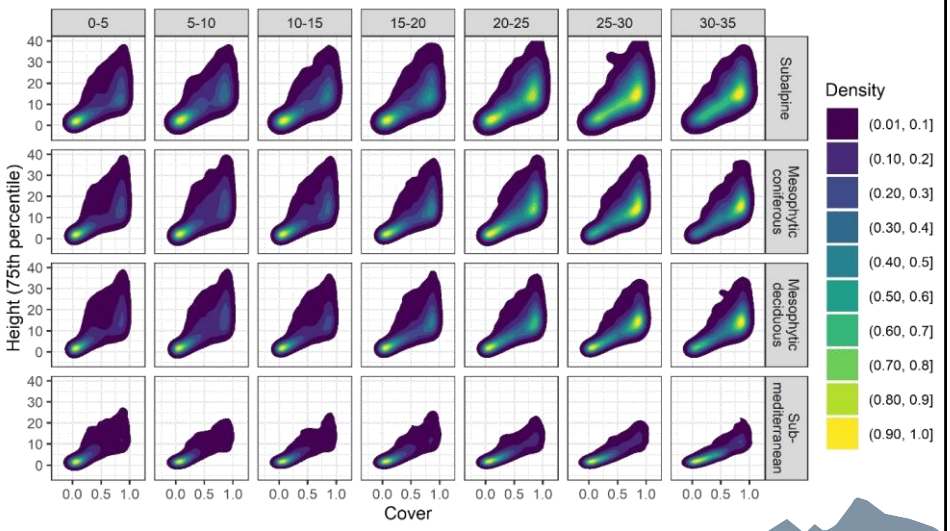

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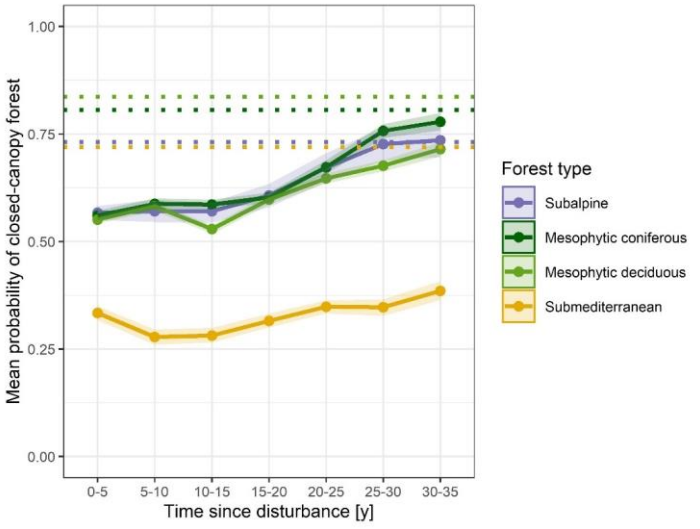

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Take home messages

Disturbances are increasing in Europe

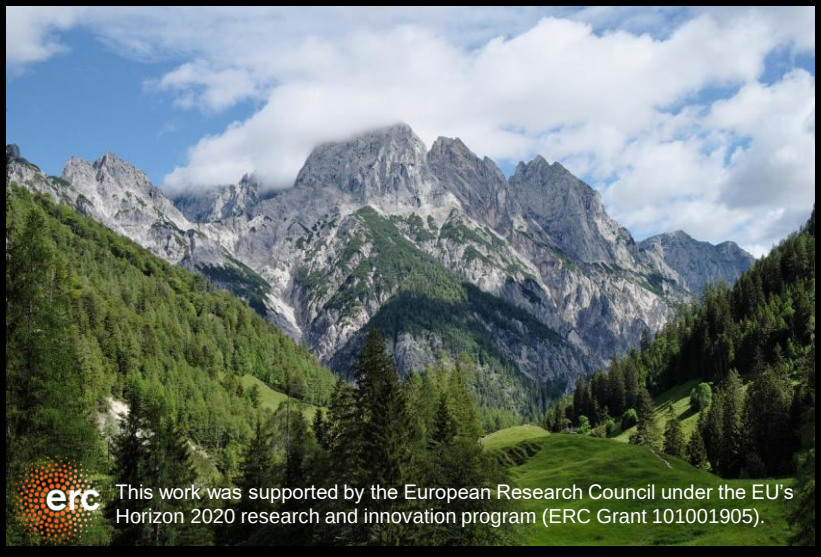
Strong increase in disturbance rates since the mid 1980s, leading to the emergence of novel forest disturbance regimes

Europe‘s forests have high resilience

The large majority of Europe‘s forests recovery relatively quickly from disturbance, indicating high resilience

Warmer & drier conditions challenge disturbance resilience

Low resilience is currently particularly evident in the Mediterranean; as climate warms, these conditions could become more widespread



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Thank you!

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